Linear Algebra and Its Application

REVIEW FOR THE FINAL EXAM

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REVIEW FOR THE FINAL EXAM

Chapter 1 Linear Equations in Linear Algebra

Chapter 2 Matrix Algebra

Chapter 3 Determinants

Chapter 4 Vector Spaces

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Chapter 7 Symmetric Matrices and Quadratic Forms



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CHAPTER 1

Linear Equations in Linear Algebra



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Chapter 1 Linear Equation in Linear Algebra

§ 1.1 Systems of Linear Equations

§ 1.2 Row Reduction and Echelon Forms

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1.1 Systems of Linear Equations

- 1. linear equation

$$a_1x_1 + a_2x_2 + \ldots + a_nx_n = b$$

Systems of Linear Equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \qquad\qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

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§ 1.1 Systems of Linear Equations

- Confficient matrix and augmented matrix

Coefficient matrix

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

augmented matrix

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

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§ 1.1 Systems of Linear Equations

- A **solution** to a system of equations

  - A system of linear equations has either

    1. No solution, or ⇒ **inconsistent**

    2. Exactly one solution, or
    3. Infinitely many solutions.

    } ⇒ **consistent**

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§ 1.1 Systems of Linear Equations

- Solving a Linear System
  - Elementary Row Operations

    1. (Replacement) Replace one row by the sum of itself and a multiple of another row.

    2. (Interchange) Interchange two rows.

    3. (Scaling) Multiply all entries in a row by a nonzero constant.

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Examples

- 1. Solving a Linear System

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ -4x_1 + 5x_2 + 9x_3 &= -9 \end{aligned}$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{bmatrix}$$

- 2. Discuss the solution of a linear system which has unknown variable

讨论线性方程组 $\begin{cases} x_1 + x_2 + 2x_3 = 2 \\ \lambda x_1 + x_2 + x_3 = 1 \\ x_1 + \lambda x_2 + x_3 = \lambda \end{cases}$，问 λ 取何值时方程组有唯一解、

无解、无穷多解？并求出方程组有无穷多解时通解的表示形式。

§ 1.1 Systems of Linear Equations

- Existence and Uniqueness Questions
  - Two fundamental questions about a linear system

1. Is the system consistent; that is, does at least one solution exist?

2. If a solution exists, is it the only one; that is, is the solution unique?

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§ 1.2 Row Reduction and Echelon Forms

- The following matrices are in echelon form:

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

pivot position

- The following matrices are in reduced echelon form:

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

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§ 1.2 Row Reduction and Echelon Forms

- Theorem 1 Uniqueness of the Reduced Echelon Form

  Each matrix is row equivalent to one and only one reduced echelon matrix.

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§ 1.2 Row Reduction and Echelon Forms

- The Row Reduction Algorithm

Step1 Begin with the leftmost nonzero column.

Step2 Select a nonzero entry in the pivot column as a pivot.

Step3 Use row replacement operations to create zeros in all positions below the pivot.

Step4 Apply steps 1-3 to the submatrix that remains. Repeat the process until there are no more nonzero rows to modify.

Step5 Beginning with the rightmost pivot and working upward and to the left, create zeros above each pivot.

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§ 1.2 Row Reduction and Echelon Forms

- Solution of Linear Systems (Using Row Reduction)

eg. Find the general solution of the following linear system

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 6 & 2 & -5 & 0 & 10 \\ 0 & 0 & 2 & -8 & 0 & 10 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 6 & 2 & -5 & 0 & 10 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

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§ 1.2 Row Reduction and Echelon Forms

The associated system now is

$$\begin{aligned} x_1 + 6x_2 \quad + 3x_4 \quad &= 0 \\ x_3 - 4x_4 \quad &= 5 \\ x_5 &= 7 \end{aligned}$$

The general solution is:

$$\begin{cases} x_1 = -6x_2 - 3x_4 \\ x_2 \text{ is free} \\ x_3 = 5 + 4x_4 \\ x_4 \text{ is free} \\ x_5 = 7 \end{cases}$$

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§ 1.2 Row Reduction and Echelon Forms

- Theorem 2 Existence and Uniqueness Theorem

A linear system is consistent if and only if the rightmost column of the augmented matrix is not a pivot column– that is , if and only if an echelon form of the augmented matrix has no row of the form

$$[0 \ \cdots \ 0 \ b] \quad b \neq 0$$

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§ 1.3 Vector Equations

- Algebraic Properties of $\mathbb{R}^n$

For all u, v, w in $\mathbb{R}^n$ and all scalars c and d:

(i) $u+v=v+u$

(ii) $(u+v)+w=u+(v+w)$

(iii) $u+0=0+u=u$

(iv) $u+(-u)=-u+u=0$

(v) $c(u+v)=cu+cv$

(vi) $(c+d)u=cu+du$

(vii) $c(du)=(cd)(u)$

(viii) $1u=u$

where –u denotes (-1)u

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§ 1.3 Vector Equations

- Subset of $\mathbb{R}^n$

  - Span $\{v_1,\ldots,v_p\}$ is collection of all vectors that can be written in the form

$$c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_p\boldsymbol{v}_p$$

with $c_1,\ldots,c_p$ scalars.

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§ 1.4 The Matrix Equation Ax = b

- 1.Definition

If A is an $m \times n$ matrix, with column $a_1,\ldots,a_n$, and if **x** is in R^n, then the product of A and **x**, denoted by $A\mathbf{x}$, is the linear combination of the columns of A using the corresponding entries in x as weights;

that is:

$$A\boldsymbol{x} = [\boldsymbol{a}_1 \ \ \boldsymbol{a}_2 \ \ \cdots \ \ \boldsymbol{a}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n$$

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§ 1.4 The Matrix Equation Ax = b

- Theorem 3

If A is an $m \times n$ matrix, with column $\mathbf{a}_1,\ldots,\mathbf{a}_n$, and if **b** is in R^m, the matrix equation

$$A\mathbf{x} = \mathbf{b}$$

has the same solution set as the vector equation

$$x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n = \boldsymbol{b}$$

which, in turn, has the same solution set as the system of linear equation whose augmented matrix is

$$[\boldsymbol{a}_1 \ \ \boldsymbol{a}_2 \ \ \cdots \ \ \boldsymbol{a}_n \ \ \boldsymbol{b}]$$

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§ 1.4 The Matrix Equation Ax = b

- **2. Existence of Solutions**

The equation $A\mathbf{x}=\mathbf{b}$ has a solution if and only if **b** is a linear combination of columns of A.

Example. Is the equation $A\mathbf{x}=\mathbf{b}$ consistent for all possible b_1, b_2, b_3?

$$A=\begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix},\ \boldsymbol{b}=\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

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§ 1.4 The Matrix Equation Ax=b

- Solution Row reduce the augmented matrix for $A\mathbf{x}=\mathbf{b}$:

$$\begin{bmatrix} 1 & 3 & 4 & b_1 \\ -4 & 2 & -6 & b_2 \\ -3 & -2 & -7 & b_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2+4b_1 \\ 0 & 7 & 5 & b_3+3b_1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2+4b_1 \\ 0 & 0 & 0 & b_3+3b_1-\frac{1}{2}(b_2+4b_1) \end{bmatrix}$$

- $\because \quad b_3+3b_1-\frac{1}{2}(b_2+4b_1) = b_1-\frac{1}{2}b_2+b_3 \neq 0$ (for some choices of **b**)

$\therefore$ The equation $A\mathbf{x}=\mathbf{b}$ is not consistent for every **b.**

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§ 1.4 The Matrix Equation Ax=b

- Theorem 4

Let A be an $m \times n$ matrix. Then the following statements are logically equivalent. That is, for a particular A, either they are all true statements or they are all false.

a. For each b in R^m, the equation $A\mathbf{x} = b$ has a solution.

b. Each b in R^m is a linear combination of the columns of A.

c. The columns of A span R^m.

d. A has a pivot position in every row.

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§ 1.4 The Matrix Equation Ax=b

- 3. Computation of *A***x**

Example . Compute Ax, where $A=\begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix}$, $\boldsymbol{x}=\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Solution.

$$\begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=x_1\begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}+x_2\begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix}+x_3\begin{bmatrix} 4 \\ -3 \\ 8 \end{bmatrix}$$

$$=\begin{bmatrix} 2x_1 \\ -x_1 \\ 6x_1 \end{bmatrix}+\begin{bmatrix} 3x_2 \\ 5x_2 \\ -2x_2 \end{bmatrix}+\begin{bmatrix} 4x_3 \\ -3x_3 \\ 8x_3 \end{bmatrix}=\begin{bmatrix} 2x_1+3x_2+4x_3 \\ -x_1+5x_2-3x_3 \\ 6x_1-2x_2+8x_3 \end{bmatrix}$$

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§ 1.4 The Matrix Equation Ax=b

- 4. Properties of the Matrix-Vector Product Ax

Theorem 5

If A is an $m \times n$ matrix, u and v are vectors in R^n, and c is a scalar, then:

a. $A(\boldsymbol{u}+\boldsymbol{v}) = A\boldsymbol{u} + A\boldsymbol{v}$.

b. $A(c\boldsymbol{u}) = c(A\boldsymbol{u})$.

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§ 1.5 Solution Set of Linear Systems

- 1. Solution of Homogeneous Linear Systems
- 2. Solution of Nonhomogeneous Systems

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§ 1.5 Solution Set of Linear Systems

- 1. Homogeneous Linear Systems $Ax = 0$
  - trivial solution (平凡解)
  - nontrivial solution (非平凡解)

- The homogeneous equation $Ax = 0$ has a nontrivial solution if and only if the equation has at least one free variable.

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§ 1.5 Solution Set of Linear Systems

- Example Solve the Homogeneous Linear Systems

$$\begin{aligned} 3x_1 + 5x_2 - 4x_3 &= 0 \\ -3x_1 - 2x_2 + 4x_3 &= 0 \\ 6x_1 + x_2 - 8x_3 &= 0 \end{aligned}$$

- Solution (1) Row reduction

$$\begin{bmatrix} 3 & 5 & -4 & 0 \\ -3 & -2 & 4 & 0 \\ 6 & 1 & -8 & 0 \end{bmatrix} \sim \begin{bmatrix} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & -9 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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Example

- (2) Row reduction to reduced echelon form

$$\begin{bmatrix} 1 & 0 & -\frac{4}{3} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{aligned} x_1 \quad -\frac{4}{3}x_3 &= 0 \\ x_2 \qquad\quad\ \ &= 0 \\ 0 \quad &= 0 \end{aligned}$$

- (3) The general solution

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{4}{3}x_3 \\ 0 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix} = x_3 v, \qquad v = \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix}$$

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§ 1.5 Solution Set of Linear Systems

- 2. Solution of Nonhomogeneous Systems

eg. Describe all solutions of Ax=b, where

$$A=\begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix},\quad \boldsymbol{b}=\begin{bmatrix} 7 \\ -1 \\ -4 \end{bmatrix}$$

Solution

$$\begin{bmatrix} 3 & 5 & -4 & 7 \\ -3 & -2 & 4 & -1 \\ 6 & 1 & -8 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{4}{3} & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{aligned} x_1 \quad & -\tfrac{4}{3}x_3 = -1 \\ x_2 \quad & = 2 \\ 0 \quad & = 0 \end{aligned}$$

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§ 1.5 Solution Set of Linear Systems

The general solution of Ax=b has the form

$$\boldsymbol{x}=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}-1+\frac{4}{3}x_3\\2\\x_3\end{bmatrix}=\begin{bmatrix}-1\\2\\0\end{bmatrix}+\begin{bmatrix}\frac{4}{3}x_3\\0\\x_3\end{bmatrix}=\begin{bmatrix}-1\\2\\0\end{bmatrix}+x_3\begin{bmatrix}\frac{4}{3}\\0\\1\end{bmatrix}$$

$\boldsymbol{p}$ $\boldsymbol{v}$

The solution set of Ax=b in parametric vector form

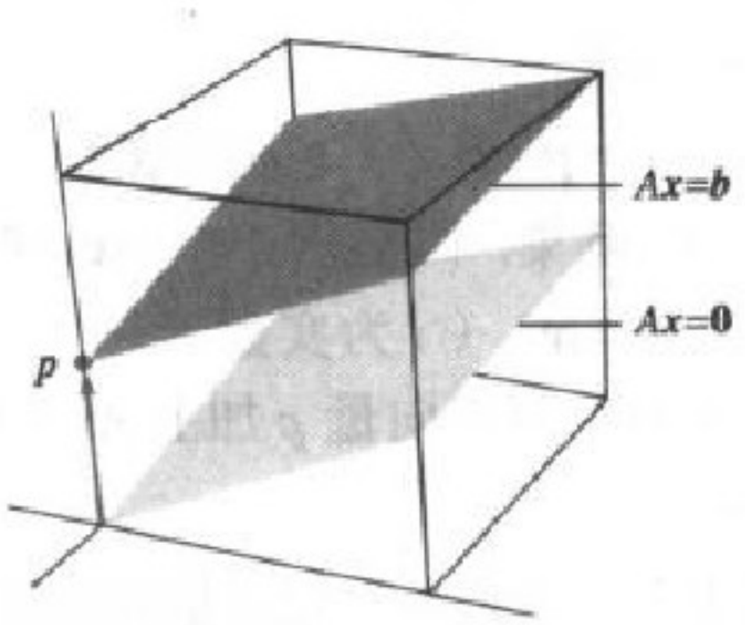
$$\boldsymbol{x}=\boldsymbol{p}+t\boldsymbol{v}$$

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§ 1.5 Solution Set of Linear Systems

- Theorem 6

  Suppose the equation Ax=b is consistent for some given b, and let p be a solution. Then the solution set of Ax=b is the set of all vectors of the form w= p + v_h, where v_h is any solution of the homogeneous equation Ax=0.



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§ 1.7 Linear Independence

- 1. Definition

  - Linear Independence

    An indexed set of vectors $\{v_1,\ldots,v_p\}$ in R^n is said to be linearly independent if the vector equation

$$x_1v_1 + x_2v_2 + \cdots + x_pv_p = 0$$

has only the trivial solution.

  - Linear Dependence

The set $\{v_1,\ldots,v_p\}$ is said to be linearly dependent if there exist weights $c_1,\ldots,c_p$, not all zero, such that

$$c_1v_1 + c_2v_2 + \cdots + c_pv_p = 0$$

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§ 1.7 Linear Independence

- Example：

$$\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \boldsymbol{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad \boldsymbol{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

a. Determine if the set $\{v_1, v_2, v_3\}$ is linearly independent.

b. If possible, find a linear dependence relation among v_1, v_2, v_3.

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Example

- a. Row reduce the augmented matrix

$$\begin{bmatrix} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Clearly, x_1 and x_2 are basic variables, and x_3 is free.

Each nonzero value of x_3 determines a nontrivial solution.

Hence $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent.

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Example 1

- b. completely row reduce the augmented matrix:

$$\begin{bmatrix} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 \quad & -2x_3 = 0 \\ x_2 + \; & x_3 = 0 \\ & 0 = 0 \end{aligned}$$

Thus, $x_1=2x_3$, $x_2=-x_3$, and x_3 is free.

Choose $x_3=5$, Then $x_1=10$ and $x_2=-5$.

So one possible linear dependence relations among v_1,v_2,v_3 is

$$10v_1 - 5v_2 + 5v_3 = 0$$

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§ 1.7 Linear Independence

The condition of linear independence:

- For Matrix Columns

  - if and only if the equation $A\mathbf{x}=0$ has only the trivial solution.

- For Sets of One or Two Vectors

  - if and only if neither of the vectors is a multiple of the other.

- For Sets of Two or More Vectors

  - Theorem 7 (Characterization of Linearly Dependent Sets)

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4. Linear Independence of Sets of Two or More Vectors

Theorem 7 (Characterization of Linearly Dependent Sets)

An indexed set $S = \{v_1,\ldots,v_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others. In fact, if S is linearly dependent and $v_1 \neq 0$, then some v_j is a linear combination of the preceding vectors, $v_1,\ldots,v_{j-1}$.

b

§ 1.7 Linear Independence

- Theorem 8

> If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set $\{v_1,\ldots,v_p\}$ in R^n is linearly dependent if $p > n$.

- Theorem 9

> If a set $S = \{v_1,\ldots,v_p\}$ in R^n contains the zero vector, then the set is linearly dependent.

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Example

- Determine by inspection if the given set is linearly dependent

a. $\begin{bmatrix}1\\7\\6\end{bmatrix}, \begin{bmatrix}2\\0\\9\end{bmatrix}, \begin{bmatrix}3\\1\\5\end{bmatrix}, \begin{bmatrix}4\\1\\8\end{bmatrix}$

b. $\begin{bmatrix}2\\3\\5\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\8\end{bmatrix}$

c. $\begin{bmatrix}-2\\4\\6\\10\end{bmatrix}, \begin{bmatrix}3\\-6\\-9\\15\end{bmatrix}$

Solution

a. ∵ The set contains 4 vectors, each has 3 entries. ∴ Dependent

b. ∵ The zero vector is in the set ∴ Dependent

c. ∵ Neither is a multiple of the other ∴ Independent

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§ 1.8 Linear Transformations

- Linear Transformations

Definition A transformation T is linear if:

(a) $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all **u**, **v** in the domain of T;

(b) $T(c\mathbf{u}) = cT(\mathbf{u})$ for all **u** and all scalars c.

If T is a linear transformation, then $\boldsymbol{T}(\mathbf{0}) = \mathbf{0}$

and for all **u**, **v** and scalars c, d: $\boldsymbol{T}(c\mathbf{u}+d\mathbf{v}) = c\boldsymbol{T}(\mathbf{u}) + d\boldsymbol{T}(\mathbf{v})$

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§ 1.9 The Matrix of A Linear Transformation

- Theorem 10

Let T: $R^n \rightarrow R^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } R^n$$

In fact, A is the $m \times n$ matrix whose j th column is the vector $T(e_j)$, where e_j is the j th column of the identity matrix in R^n:

$$A = [T(e_1) \ldots T(e_n)]$$

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§ 1.9 The Matrix of A Linear Transformation

- Example: Find the standard matrix A for the dilation transformation $T(\mathbf{x}) = 3\mathbf{x}$, for $\mathbf{x}$ in R^2

Solution:

$$T(\boldsymbol{e}_1) = 3\boldsymbol{e}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \text{ 和 } T(\boldsymbol{e}_2) = 3\boldsymbol{e}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

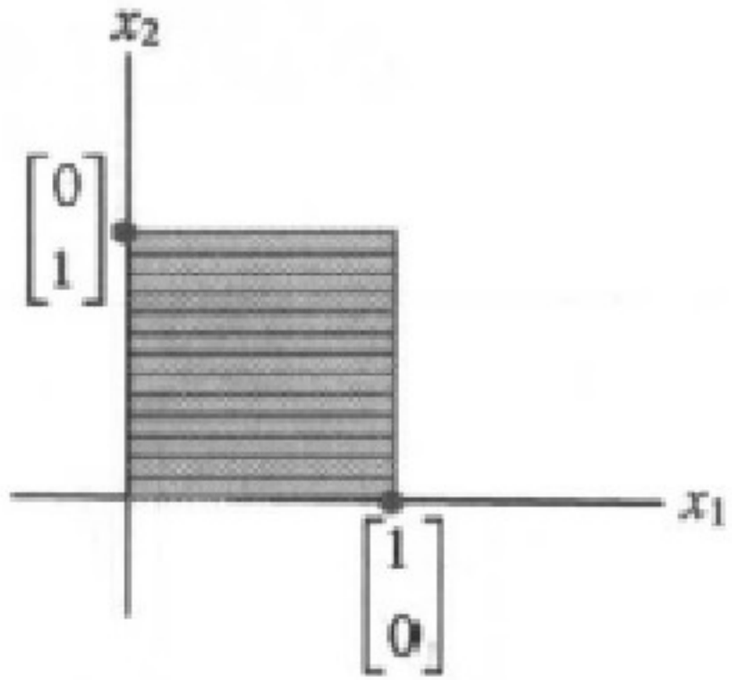
$$A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

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§ 1.9 The Matrix of A Linear Transformation

- 2. Geometric Linear Transformations of R^2

会求线性变换的标准矩阵，不要求记住



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表 1-2　对称

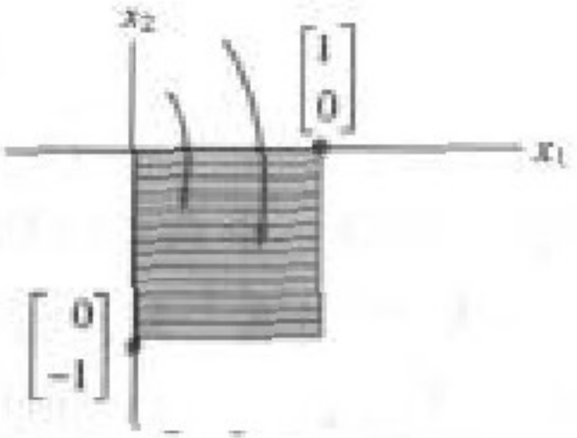
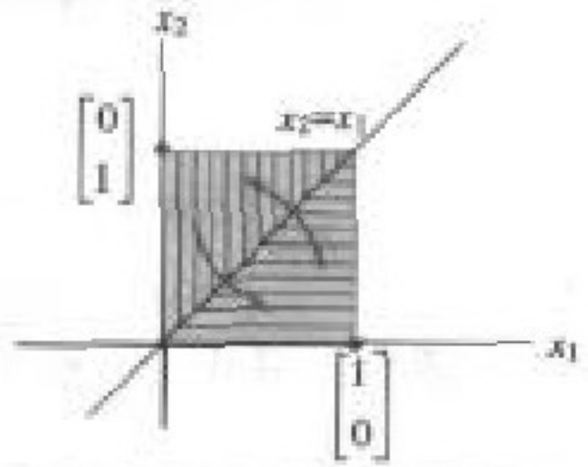
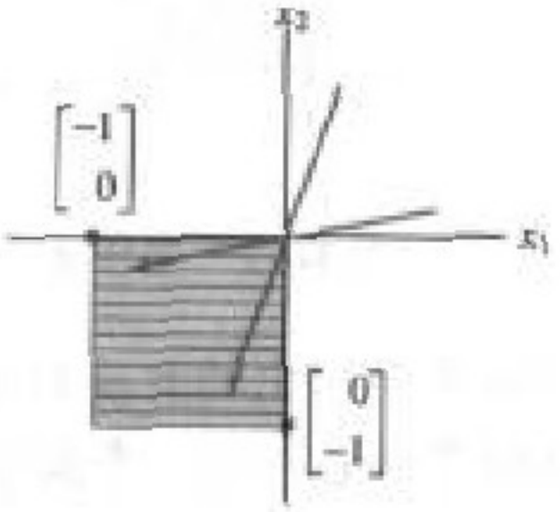
变　　换	单位正方形的像	标准矩阵
关于 x_1 轴的对称		$\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$
关于直线 $x_2 = x_1$ 的对称		$\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$
关于原点的对称		$\begin{bmatrix}-1 & 0\\0 & -1\end{bmatrix}$

表 1-3　收缩与拉伸

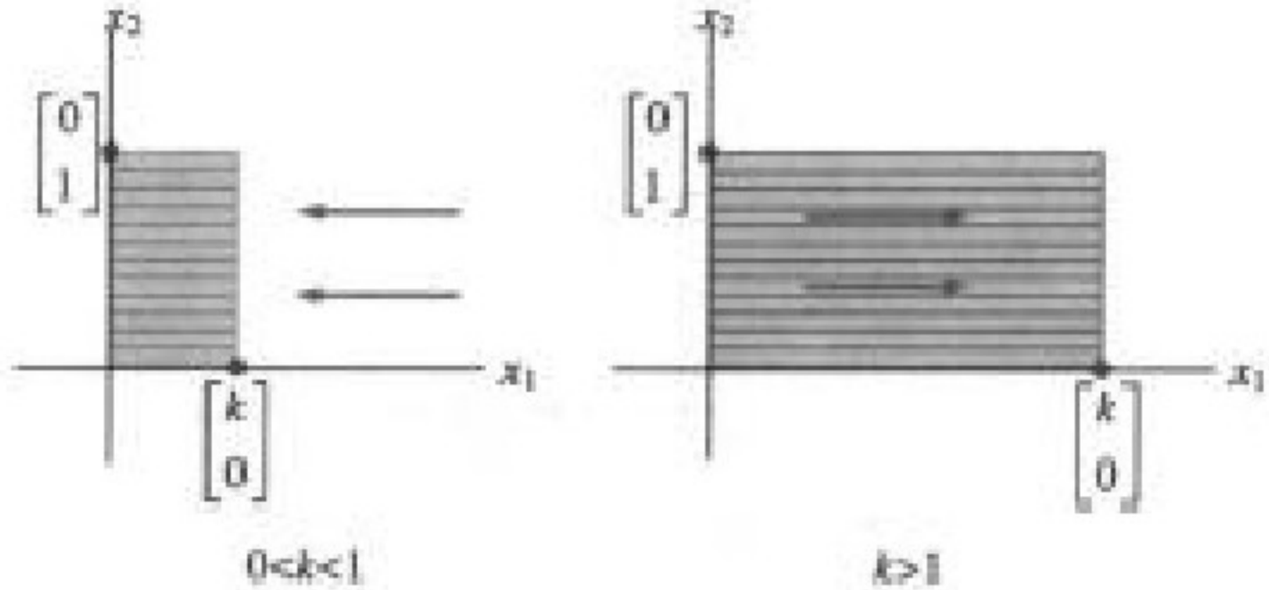
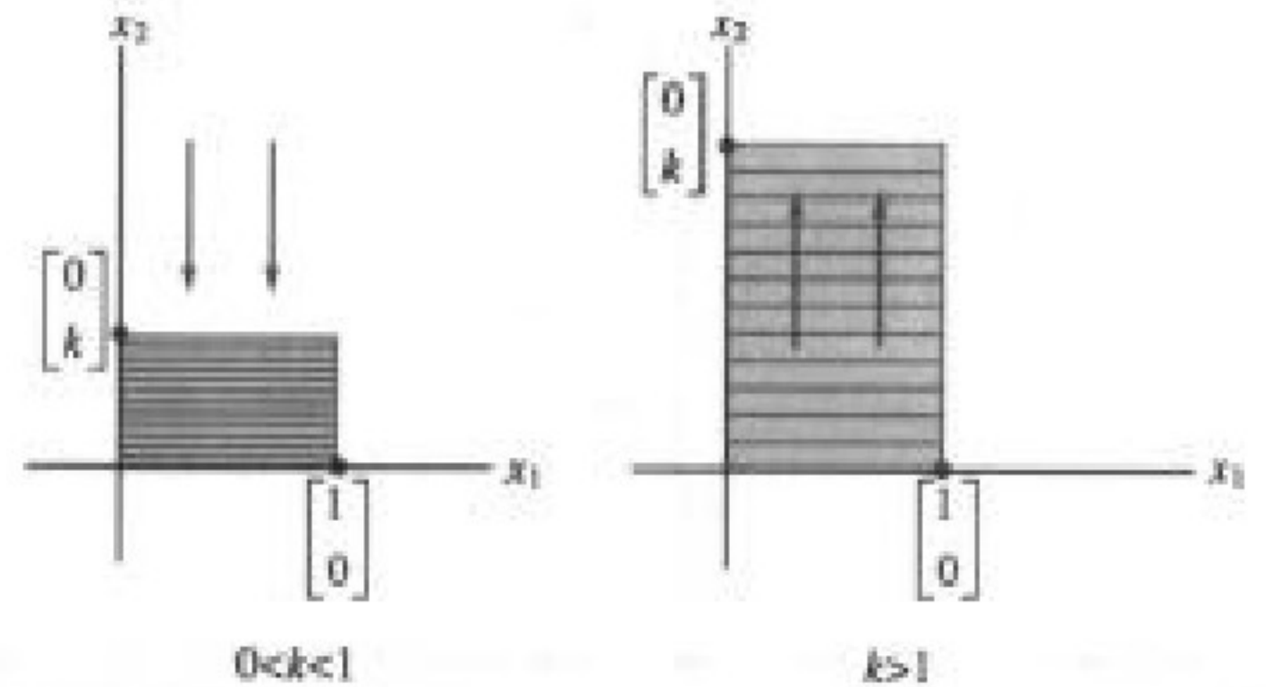
变　　换	单位正方形的像	标准矩阵
水平收缩与拉伸		$\begin{bmatrix}k & 0\\0 & 1\end{bmatrix}$
垂直收缩与拉伸		$\begin{bmatrix}1 & 0\\0 & k\end{bmatrix}$

表 1-4　剪切变换

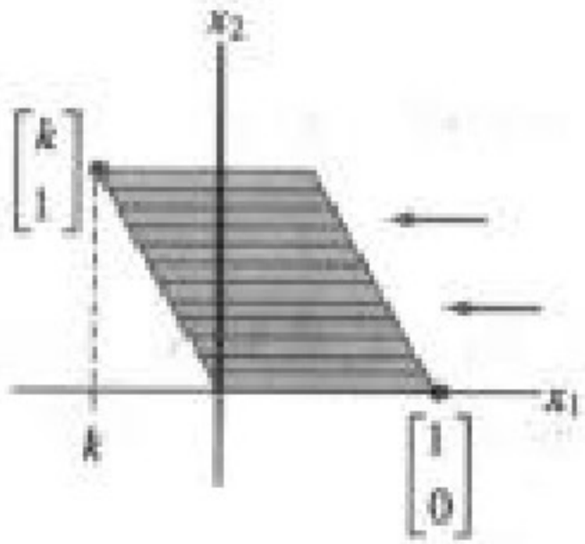
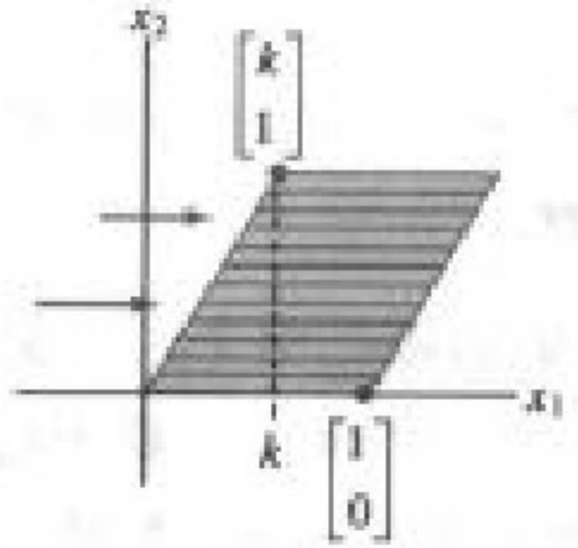
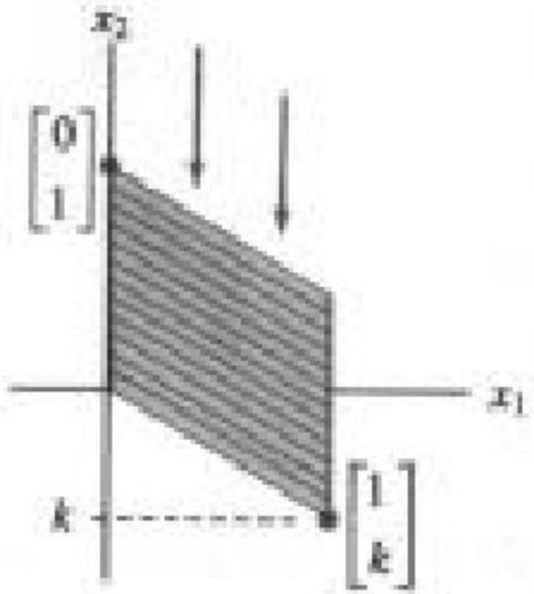
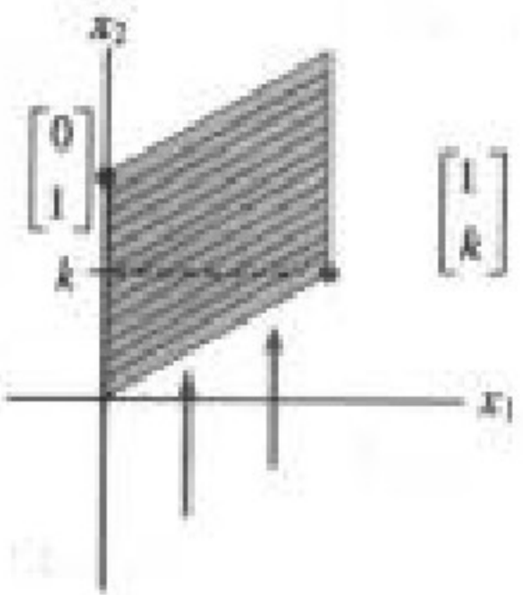
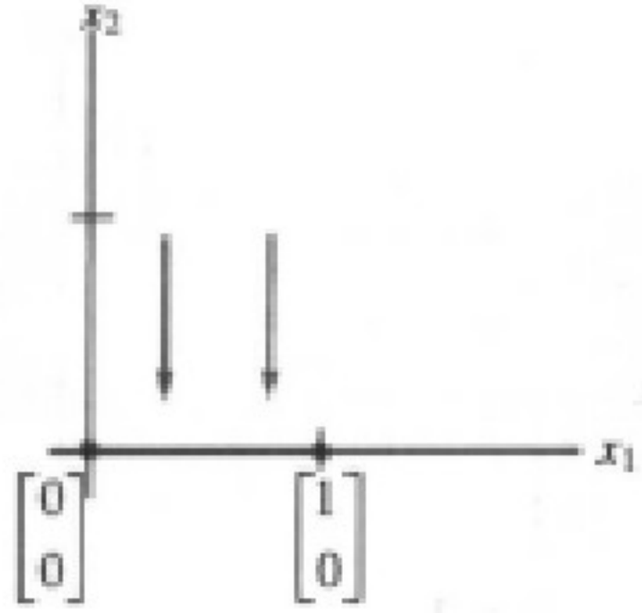
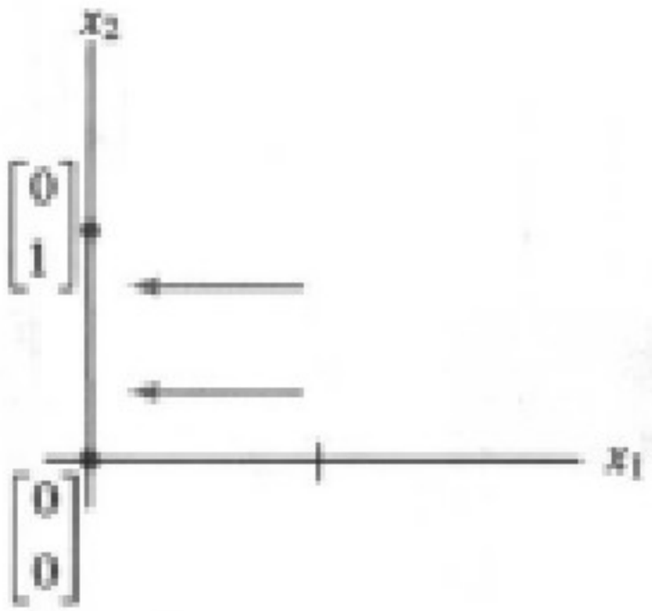
变　　换	单位正方形的像	标准矩阵
水平剪切	 $k<0$  $k>0$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$
垂直剪切	 $k<0$  $k>0$	$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$

表 1-5　投影

变　　换	单位正方形的像	标准矩阵
投影到 x_1 轴上		$\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$
投影到 x_2 轴上		$\begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}$

§ 1.9 The Matrix of A Linear Transformation

3. Existence and Uniqueness Questions

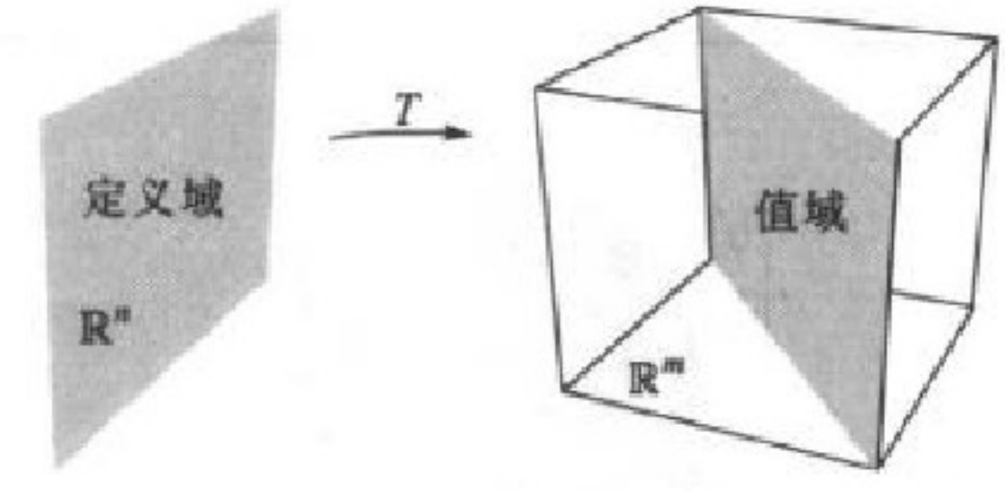
Definition

(1) A mapping T : R^n->R^m is said to be **onto** R^m if each **b** in R^m is the image of at least one **x** in R^n (满射)

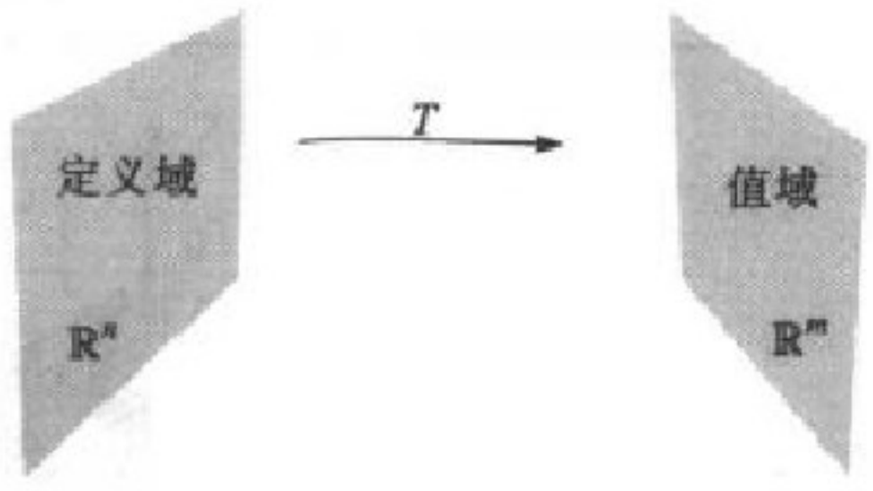
(2) A mapping T : R^n->R^m is said to be **one-to-one** if each **b** in R^m is the image of at most one **x** in R^n (单射)

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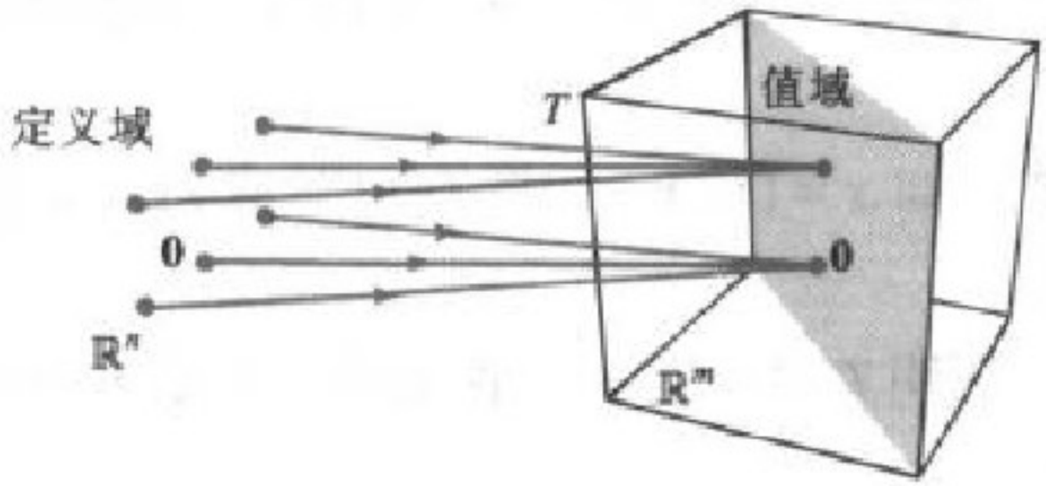
§ 1.9 The Matrix of A Linear Transformation



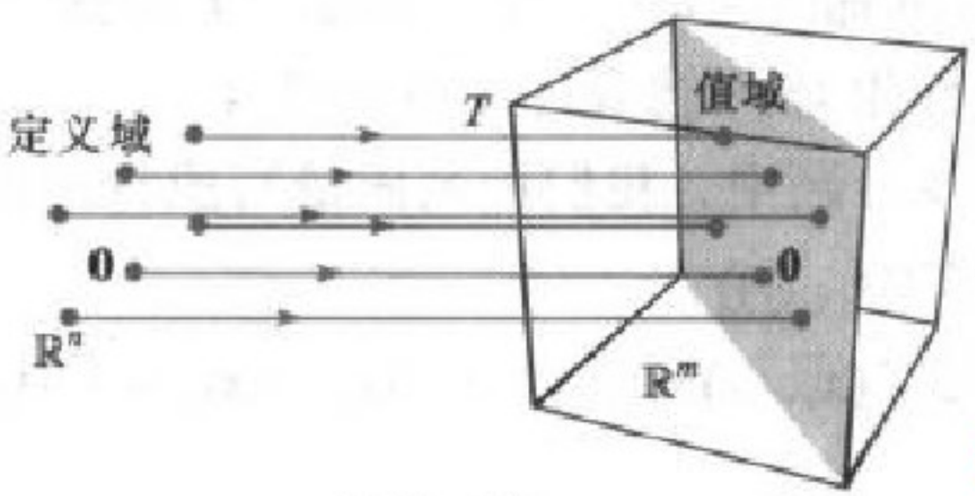
T不是到$\mathbb{R}^m$上的



T是到$\mathbb{R}^m$上的



T不是一对一



T是一对一

§ 1.9 The Matrix of A Linear Transformation

- Theorem 11

Let T: R^n->R^m be a linear transformation. Then T is one-to-one if and only if the equation $T(\mathbf{x}) = 0$ has only the trivial solution.

- Theorem 12

Let T: R^n->R^m be a linear transformation and let A be the standard matrix for T. Then:

a. T maps R^n onto R^m if and only if the columns of A span R^m

b. T is one-to-one if and only if the columns of A are linearly independent

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Chapter 2 Matrix Algebra

§ 2.1 Matrix Operation

§ 2.2 The Inverse of a Matrix

§ 2.3 Characterizations of Invertible Matrices

§ 2.4 Partitioned Matrices

§ 2.5 Matrix Factorizations



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§ 2.1 Matrix Operation

- Sum

  A + B: the sum of the corresponding entries in A and B

- Scalar Multiples

  *c*A : the multiples *c* of all the entries in A

The sum A + B is defined only when A and B are the same size.

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§ 2.1 Matrix Operation

- Theorem 1

Let A, B and C be matrices of the same size, and let r and s be scalars.

a. $A+B=B+A$

b. $(A+B)+C=A+(B+C)$

c. $A+0=A$

d. $r(A+B)=rA+rB$

e. $(r+s)A=rA+sA$

f. $r(sA)=(rs)A$

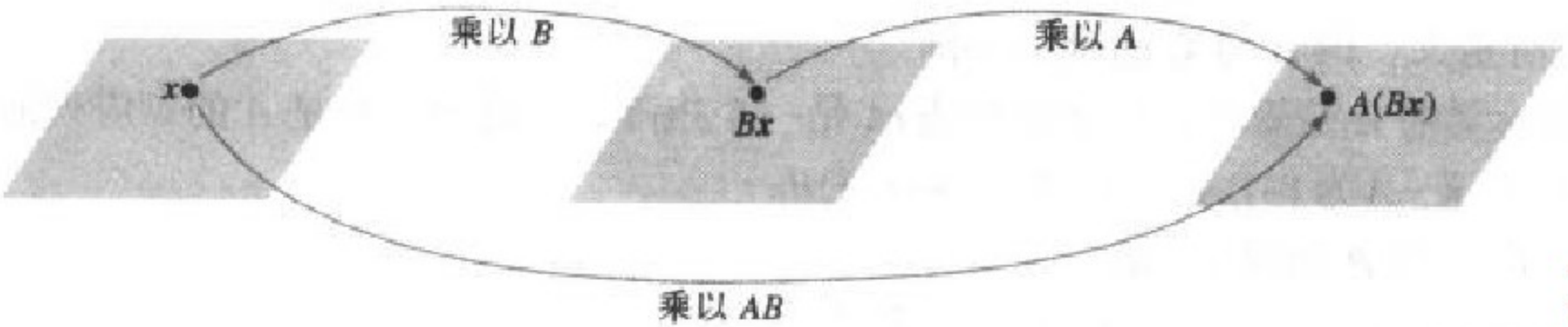
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§ 2.1 Matrix Operation

- 2. Matrix Multiplication



$$A(Bx) = (AB)x$$



§ 2.1 Matrix Operation

2. Matrix Multiplication

If A is an $m \times n$ matrix, and if B is an $n \times p$ matrix with columns $b_1,\ldots,b_p$, then the product *AB* is the $m \times p$ matrix whose columns are A$b1,\ldots,$Ab_p.

That is

$$AB = A[b_1 \quad b_2 \quad \cdots \quad b_p] = [Ab_1 \quad Ab_2 \quad \cdots \quad Ab_p]$$

Each column of AB is a linear combination of the columns of A using weights from the corresponding column of B.

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§ 2.1 Matrix Operation

- Example : Compute AB, where $A=\begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}$, $B=\begin{bmatrix}4 & 3 & 6\\ 1 & -2 & 3\end{bmatrix}$

  Solution Write $B=[\boldsymbol{b}_1 \ \ \boldsymbol{b}_2 \ \ \boldsymbol{b}_3]$, and compute:

$$\boldsymbol{Ab}_1=\begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}\begin{bmatrix}4\\ 1\end{bmatrix},\quad \boldsymbol{Ab}_2=\begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}\begin{bmatrix}3\\ -2\end{bmatrix},\quad \boldsymbol{Ab}_3=\begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}\begin{bmatrix}6\\ 3\end{bmatrix}$$

$$=\begin{bmatrix}11\\ -1\end{bmatrix} \qquad =\begin{bmatrix}0\\ 13\end{bmatrix} \qquad =\begin{bmatrix}21\\ -9\end{bmatrix}$$

$$\boldsymbol{AB}=\boldsymbol{A}[\boldsymbol{b}_1 \ \ \boldsymbol{b}_2 \ \ \boldsymbol{b}_3]=\begin{bmatrix}11 & 0 & 21\\ -1 & 13 & -9\end{bmatrix}$$

$\boldsymbol{Ab}_1 \ \boldsymbol{Ab}_2 \ \boldsymbol{Ab}_3$

§ 2.1 Matrix Operation

Row- Column Rule for Computing AB

If the product *AB* is defined, then the entry in row i and column *j* of *AB* is the sum of the product of corresponding entries from row *i* of *A* and column *j* of *B*. If $(AB)_{ij}$ denotes the (i, j)-entry in *AB*, and if *A* is an $m \times n$ matrix, then

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

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§ 2.1 Matrix Operation

3. Properties of Matrix Multiplication

Theorem 2

Let A be an m×n matrix, and let B and C have sizes for which the indicated sums and products are defined.

a. $A(BC)=(AB)C$ （乘法结合律）

b. $A(B+C)=AB+AC$ （乘法左分配律）

c. $(B+C)A=BA+CA$ （乘法右分配律）

d. $r(AB)=(rA)B=A(rB)$， r 为任意数

e. $I_mA=A=AI_n$ （矩阵乘法的恒等式）

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§ 2.1 Matrix Operation

- Warnings:

1. In general, AB ≠ BA.

2. The cancellation laws do not hold for matrix multiplication. That is, if AB =AC, then it is not true in general that B = C.

3. If a product AB is the zero matrix, you cannot conclude in general that either A = 0 or B = 0.

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§ 2.1 Matrix Operation

- 4. Power of a Matrix

If A is an $m \times n$ matrix and if k is a positive integer, then A_k denotes the product of k copies of A:

$$A^k = \underbrace{A \cdots A}_{k\text{个}}$$

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§ 2.1 Matrix Operation

Theorem 3

Let A and B denote matrices, whose sizes are appropriate for the following sums and products.

a. $(A^T)^T = A$.

b. $(A+B)^T = A^T + B^T$.

c. 对任意数 r， $(rA)^T = rA^T$.

d. $(AB)^T = B^T A^T$.

The transpose of a product of matrices equals the product of their transposes in the reverse order.

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§ 2.2 The Inverse of a Matrix

- 1. The Inverse of a Matrix

For matrix: An n×n matrix A is said to be invertible if there is an n×n matrix C such that $CA = I$ and $AC = I$. ($I = I_n$)

singular matrix: not invertible matrix

nonsigular matrix: invertible matrix

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§ 2.2 The Inverse of a Matrix

Theorem 5

If A is an invertible $n \times n$ matrix, then for each b in R^n, the equation $Ax = b$ has the unique solution $x = A^{-1}b$.

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§ 2.2 The Inverse of a Matrix

- Example : Use the inverse of the matrix $A=\begin{bmatrix}3 & 4\\ 5 & 6\end{bmatrix}$ to solve the system.

$$\begin{aligned}3x_1+4x_2&=3\\ 5x_1+6x_2&=7\end{aligned}$$

Solution : This system is equivalent to Ax = b, so

$$\boldsymbol{x}=A^{-1}\boldsymbol{b}=\begin{bmatrix}-3 & 2\\ 5/2 & -3/2\end{bmatrix}\begin{bmatrix}3\\ 7\end{bmatrix}=\begin{bmatrix}5\\ -3\end{bmatrix}$$

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§ 2.2 The Inverse of a Matrix

Theorem 6

a. If A is an invertible matrix, then A^{-1} is invertible and

$$(A^{-1})^{-1} = A$$

b. **If A and B are n×n invertible matrices, then so is AB, and the inverse of AB is the product of the inverses of A and B in the reverse order. That is,**

$$(AB)^{-1} = B^{-1}A^{-1}$$

c. If A is an invertible matrix, then so is A^T, and the inverse of A^T is the transpose of A^{-1}. That is,

$$(A^T)^{-1} = (A^{-1})^T$$

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§ 2.2 The Inverse of a Matrix

- 3. An Algorithm for Finding A^{-1}

Algorithm for Finding A^{-1}

Row reduce the augmented matrix [A I]. If A is row equivalent to I, then [A I] is row equivalent to [I A^{-1}]. Otherwise, A does not have an inverse.

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§ 2.2 The Inverse of a Matrix

- Example : Find the inverse of the matrix A, if it exists.

$$A=\begin{bmatrix}0 & 1 & 2\\ 1 & 0 & 3\\ 4 & -3 & 8\end{bmatrix}$$

Solution:

$$[A\ \ I]=\begin{bmatrix}0 & 1 & 2 & 1 & 0 & 0\\ 1 & 0 & 3 & 0 & 1 & 0\\ 4 & -3 & 8 & 0 & 0 & 1\end{bmatrix}\sim\begin{bmatrix}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 4 & -3 & 8 & 0 & 0 & 1\end{bmatrix}$$

$$\sim\begin{bmatrix}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 0 & -3 & -4 & 0 & -4 & 1\end{bmatrix}\sim\begin{bmatrix}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 0 & 0 & 2 & 3 & -4 & 1\end{bmatrix}$$

$$\sim\begin{bmatrix}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 0 & 0 & 1 & 3/2 & -2 & 1/2\end{bmatrix}\sim\begin{bmatrix}1 & 0 & 0 & -9/2 & 7 & -3/2\\ 0 & 1 & 0 & -2 & 4 & -1\\ 0 & 0 & 1 & 3/2 & -2 & 1/2\end{bmatrix}$$

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§ 2.2 The Inverse of a Matrix

Solution:

Since A ~ *I*, we conclude that A is invertible by Theorem 7 and

$$A^{-1} = \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix}$$

To check the final answer:

$$AA^{-1} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix} \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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§ 2.3 Characterizations of Invertible Matrices

Theorem 8 The Invertible Matrix Theorem

Let A be a square n×n matrix. Then the following statements are equivalent.

That is, for a given A, the statements are either all true or all false.

a. A is an invertible matrix.

b. A is row equivalent to the n×n identity matrix.

c. A has n pivot positions.

d. The equation Ax=0 has only the trivial solution.

e. The columns of A form a linearly independent set.

f. The linear transformation $x \to Ax$ is one-to-one.

g. The equation Ax=b has at least one solution for each b in R^n

h. The columns of A span R^n.

i.The linear transformation $x \to Ax$ maps R^n onto R^n.

j. There is an n×n matrix C such that CA = I.

k. There is an n×n matrix D such that AD = I.

l. A^T is an invertible matrix.

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§ 2.4 Partitioned Matrices

- 1. Partitions of Matrices
- 2. Addition and Scalar Multiplication
- 3. Multiplication of Partitioned Matrices
- 4. Inverses of Partitioned Matrices

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§ 2.4 Partitioned Matrices

- 1. Partitions of Matrices
- Example : The matrix

$$A=\left[\begin{array}{ccc|cc|c} 3 & 0 & -1 & 5 & 9 & -2 \\ -5 & 2 & 4 & 0 & -3 & 1 \\ \hline -8 & -6 & 3 & 1 & 7 & -4 \end{array}\right]$$

2 × 3 partitioned matrix

$$A=\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

where

$$A_{11}=\begin{bmatrix} 3 & 0 & -1 \\ -5 & 2 & 4 \end{bmatrix}, \quad A_{12}=\begin{bmatrix} 5 & 9 \\ 0 & -3 \end{bmatrix}, \quad A_{13}=\begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$A_{21}=[-8 \quad -6 \quad 3], \quad A_{22}=[1 \quad 7], \quad A_{23}=[-4]$$

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§ 2.4 Partitioned Matrices

- 2. Addition and Scalar Multiplication

A and B : Matrices of same size and partitioned in the same way

A + B: the same partition of the ordinary matrix sum A + B.

each block is the sum of corresponding blocks of A and B.

*c*A: Multiplication of a partitioned matrix A by a scalar *c*

computed block by block.

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§ 2.4 Partitioned Matrices

3. Multiplication of Partitioned Matrices: AB

Theorem 10 Column-Row Expansion of AB

If A is m×n matrix and B is n×p, then

$$AB = \begin{bmatrix} \mathrm{col}_1(A) & \mathrm{col}_2(A) & \cdots & \mathrm{col}_n(A) \end{bmatrix} \begin{bmatrix} \mathrm{row}_1(B) \\ \mathrm{row}_2(B) \\ \vdots \\ \mathrm{row}_n(B) \end{bmatrix}$$

$$= \mathrm{col}_1(A)\mathrm{row}_1(B) + \cdots + \mathrm{col}_n(A)\mathrm{row}_n(B)$$

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§ 2.5 Matrix Factorizations

- 1. The LU Factorization
- 2. An LU Factorization Algorithm

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§ 2.5 Matrix Factorizations

LU Factorization

A is m×n matrix and can be row reduced to echelon form without row interchanges. Then A can be written in:

$$A = LU$$

where L is m×m lower triangular matrix and U is m×n echelon form of A.

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{U}$$

Such a factorization is called an LU factorization.

L : invertible, a unit lower triangular matrix

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§ 2.5 Matrix Factorizations

- Example： Find an LU factorization of

$$A=\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

- Solution：

Since A has four rows, L should be 4×4

$$L=\begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & & 1 & 0 \\ -3 & & & 1 \end{bmatrix}$$

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§ 2.5 Matrix Factorizations

- Row reduction of A to an echelon form U:

$$A=\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix} = A_1$$

$$\sim A_2=\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix} = U$$

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§ 2.5 Matrix Factorizations

- At each pivot column, divide the highlighted entries by the pivot and place the result into L:

$$\begin{matrix}
\begin{bmatrix} 2 \\ -4 \\ 2 \\ -6 \end{bmatrix} & \begin{bmatrix} 3 \\ -9 \\ 12 \end{bmatrix} & \begin{bmatrix} 2 \\ 4 \end{bmatrix} & [5] \\
\div 2 & \div 3 & \div 2 & \div 5 \\
\downarrow & \downarrow & \downarrow & \downarrow
\end{matrix}$$

$$\begin{bmatrix} 1 & & & \\ -2 & 1 & & \\ 1 & -3 & 1 & \\ -3 & 4 & 2 & 1 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$

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Chapter 3 Determinants

§ 3.1 Introduction to Determinants

§ 3.2 Properties of Determinants

§ 3.3 Cramer's Rule, Volume, and Linear Transformations

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§ 3.1 Introduction to Determinants

- **Definition:**

$$\det A = a_{11} \cdot \det A_{11} - a_{12} \cdot \det A_{12} + \cdots + (-1)^{1+n} a_{1n} \cdot \det A_{1n}$$

$$= \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det A_{1j}$$

- Given A = [a_{ij}]. the (i. j)-cofactor of A is the number C_{ij} given by

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

$$\det A = a_{11} \cdot C_{11} + a_{12} \cdot C_{12} + \cdots + a_{1n} \cdot C_{1n}$$

- This formula is called a cofactor expansion across the first row of A.

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§ 3.1 Introduction to Determinants

Theorem 1

The determinant of an n × n matrix A can be computed by a cofactor expansion across any row or down any column. The expansion across the ith row using the cofactors in (4) is

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

The cofactor expansion down the j_{th} column is

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

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§ 3.1 Introduction to Determinants

- Example: Compute det A, where

$$A=\begin{bmatrix} 3 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

- Solution: Compute

$$\det A = 3\cdot\begin{bmatrix} 2 & -5 & 7 & 3 \\ 0 & 1 & 5 & 0 \\ 0 & 2 & 4 & -1 \\ 0 & 0 & -2 & 0 \end{bmatrix} - 0\cdot C_{21} + 0\cdot C_{31} - 0\cdot C_{41} + 0\cdot C_{51}$$

$$= 3\cdot 2\cdot\begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix} = 3\cdot 2\cdot(-2) = -12$$

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§ 3.1 Introduction to Determinants

Theorem 2

If A is a triangular matrix, then det A is the product of the entries on the main diagonal of A.

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§ 3.2 Properties of Determinants

1. Properties of Determinants

Theorem 3 Row Operations

Let *A* be square matrix.

a. If a multiple of one row of *A* is added to another row to produce a matrix *B*, then det *B* = det *A*.

b. If two rows of A are interchanged to produce *B*, then det *B* = -det *A*.

c. If one row of A is multiplied by k to produce *B*, then det *B* = kdet *A*.

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§ 3.2 Properties of Determinants

- Example： Compute det *A*, where

$$A=\begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}.$$

- Solution Compute

$$\det A=\begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix}=\begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ -1 & 7 & 0 \end{vmatrix}=\begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix}$$

An interchange of rows 2 and 3 reverses the sign of the determinant,

so

$$\det A=-\begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix}=-(1)(3)(-5)=15$$

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§ 3.2 Properties of Determinants

Theorem 4

A square matrix A is invertible if and only if det A ≠ 0.

Theorem 5

If A is an n×n matrix, then det A^T = det A.

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Chapter 4 Vector Spaces

- § 4.1 Vector Spaces and Subspaces
- § 4.2 Null Spaces, Column Spaces, and Linear Transformation
- § 4.3 Linearly Independent Sets; Bases
- § 4.4 Coordinate System
- § 4.5 The Dimension of a Vector Space
- § 4.6 Rank



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§ 4.1 Vector Spaces and Subspaces

1. Vector Spaces

Definition A **vector space** is a nonempty set V of objects, called *vectors*, on which are defined two operations, called *addition* and *multiplication by scalars* (real numbers), subject to the ten axioms below. The axioms must hold for all **u**, **v** and **w** in V and for all scalars c and d.

(1) $\mathbf{u} + \mathbf{v}$ is in V. (2) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$. (3) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.

(4) There is a zero vector $\mathbf{0}$ in V such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.

(5) For each **u** in V, there is vector **u** in V such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.

(6) $c\mathbf{u}$ is in V. (7) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$. (8) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$.

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§ 4.1 Vector Spaces and Subspaces

- 2. Subspaces

Definition A **subspace** of a vector space V is a subset H of V that has three properties:

a. The zero vector of V is in H^2.

b. H is closed under vector addition. That is, for each **u** and **v** are in H, **u + v** is in H.

c. H is closed under scalar multiplication. That is, for each **u** in H and each scalar c, c**u** is in H.

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§ 4.1 Vector Spaces and Subspaces

EXAMPLE: Let $H = \left\{ \begin{bmatrix} a \\ 0 \\ b \end{bmatrix} : a \text{ and } b \text{ are real} \right\}$. Show that H is a subspace of $\mathbf{R}^3$.

Solution: Verify properties a, b and c of the definition of a subspace.

a. The zero vector of $\mathbf{R}^3$ is in H (let $a =$ __0__ and $b =$ __0__).

b. Adding two vectors in H always produces another vector whose second entry is __0__ and therefore the sum of two vectors in H is also in H. (H is closed under addition)

c. Multiplying a vector in H by a scalar produces another vector in H (H is closed under scalar multiplication).

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Since properties a, b, and c hold, V is a subspace of $\mathbf{R}^3$.

§ 4.1 Vector Spaces and Subspaces

Theorem 1

If $\mathbf{v}_1,\ldots,\mathbf{v}_p$ are in a vector space V, then Span$\{\mathbf{v}_1,\ldots,\mathbf{v}_p\}$ is a subspace of V.

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§ 4.1 Vector Spaces and Subspaces

- Example 11 Let *H* be the set of all vectors of the form (a-3b, b-a, a, b), where a and b are arbitrary scalars. That is,

$$H = \{(a-3b, b-a, a, b) : a, b \in \mathbb{R}\}$$

Show that *H* is a subspace of R^4.

- Solution Write the vectors in H as column vectors. Then an arbitrary vector in H has the form:

$$\begin{bmatrix} a-3b \\ b-a \\ a \\ b \end{bmatrix} = a\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

v_1 v_2

This calculation shows that H = Span{v1,v2}.

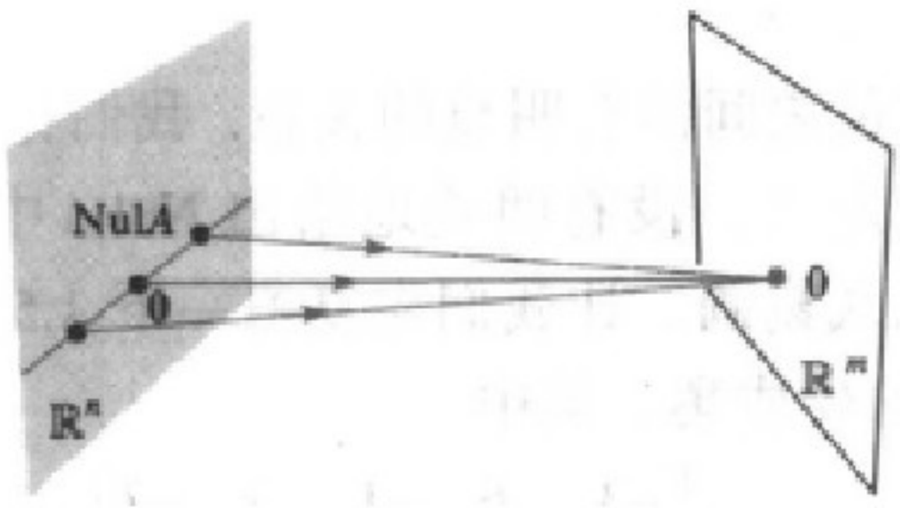
Thus *H* is a subspace of R^4

§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

- **The Null Space of a Matrix**
  - Definition:

The **null space** of an $m \times n$ matrix A, written as Nul A, is the set of all solutions to the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

$$\text{Nul } A = \{\mathbf{x} : \mathbf{x} \text{ is in } \mathbf{R}^n \text{ and } A\mathbf{x} = \mathbf{0}\} \qquad \text{(set notation)}$$



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§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

Theorem 2

The null space of an $m \times n$ matrix A is a subspace of $\mathbf{R}^n$.
Equivalently, the set of all solutions to a system $A\mathbf{x} = \mathbf{0}$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbf{R}^n$.

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§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

- A linear combination of vectors where *the weights are the free variables*. That is,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$= x_2\boldsymbol{u} + x_4\boldsymbol{v} + x_5\boldsymbol{w}$$

Every linear combination of u, v and w is an element of Nul A.

Thus {u, v, w} is a spanning set for Nul A.

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§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

- The Column Space of a Matrix

Definition：

The **column space** of an $m \times n$ matrix A (Col A) is the set of all linear combinations of the columns of A.

If $A = \begin{bmatrix} \mathbf{a}_1 & \dots & \mathbf{a}_n \end{bmatrix}$, then

$$\text{Col } A = \text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$$

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§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

Theorem 3

The column space of an $m \times n$ matrix A is a subspace of $\mathbf{R}^m$.

Recall that if $A\mathbf{x} = \mathbf{b}$, then $\mathbf{b}$ is a linear combination of the columns of A. Therefore

$$\text{Col } A = \{\mathbf{b} : \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \text{ in } \mathbf{R}^n\}$$

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§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

- Example： Find a matrix A such that W = Col A.

$$W=\left\{\begin{bmatrix}6a-b\\ a+b\\ -7a\end{bmatrix}:a,b\in\mathbb{R}\right\}$$

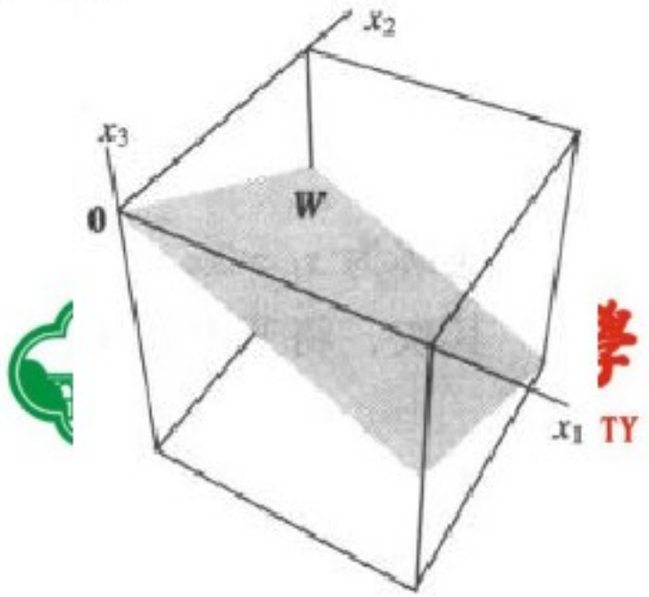
- Solution： First, write W as a set of linear combinations.

$$W=\left\{a\begin{bmatrix}6\\1\\-7\end{bmatrix}+b\begin{bmatrix}-1\\1\\0\end{bmatrix}:a,b\in\mathbb{R}\right\}=\text{Span}\left\{\begin{bmatrix}6\\1\\-7\end{bmatrix},\begin{bmatrix}-1\\1\\0\end{bmatrix}\right\}$$

Let

$$A=\begin{bmatrix}6&-1\\1&1\\-7&0\end{bmatrix}$$

Then W = Col A.



§ 4.3 Linearly Independent Sets; Bases

Linearly Independent Sets

Theorem 4

An indexed set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ of two or more vectors, with $\mathbf{v}_1 \neq \mathbf{0}$, is linearly dependent if and only if some vector $\mathbf{v}_j$ $(j > 1)$ is a linear combination of the preceding vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$.

Example Let $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ be a set of vectors in $\mathbf{P}_2$ where $\mathbf{p}_1(t) = t$, $\mathbf{p}_2(t) = t^2$, and $\mathbf{p}_3(t) = 4t + 2t^2$. Is this a linearly dependent set?

Solution: Since $\mathbf{p}_3 =$ ____**4**____ $\mathbf{p}_1 +$ ____**2**____ $\mathbf{p}_2$, $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ is a linearly _____**dependent**_____ set.

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§ 4.3 Linearly Independent Sets; Bases

- **The Basis set**

- Definition

Let H be a subspace of a vector space V. An indexed set of vectors $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_p\}$ in V is a basis for H if

(i) β is a linearly independent set, and

(ii) $H = \text{Span}\{\mathbf{b}_1, \ldots, \mathbf{b}_p\}$.

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EXAMPLE: Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$.

Is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ a **basis** for $\mathbf{R}^3$?

Solution: Again, let $A = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$. Using row reduction,

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 5 \end{bmatrix}$$

and since there are 3 pivots, the columns of A are linearly independent and they span $\mathbf{R}^3$ by the IMT. Therefore $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a **basis** for $\mathbf{R}^3$.

§ 4.3 Linearly Independent Sets; Bases

- **The Spanning Set Theorem**

THEOREM 5 The Spanning Set Theorem

Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ be a set in V and let $H = \text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$.

a. If one of the vectors in S - say $\mathbf{v}_k$ - is a linear combination of the remaining vectors in S, then the set formed from S by removing $\mathbf{v}_k$ still spans H.

b. If $H \neq \{\mathbf{0}\}$, some subset of S is a basis for H.

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§ 4.3 Linearly Independent Sets; Bases

EXAMPLE: Suppose $\mathbf{v}_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ and $\mathbf{v}_3 = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$.

Solution: If $\mathbf{x}$ is in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, then

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3(\underline{\quad 2 \quad}\mathbf{v}_1 + \underline{\quad 3 \quad}\mathbf{v}_2)$$

$$= \underline{\quad (c_1 + 2c_3) \quad}\mathbf{v}_1 + \underline{\quad (c_2 + 3c_3) \quad}\mathbf{v}_2$$

Therefore,

$$\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}.$$

§ 4.3 Linearly Independent Sets; Bases

- **Bases for Null A and Col A**

EXAMPLE: Find a basis for Nul A where

$$A = \begin{bmatrix} 3 & 6 & 6 & 3 & 9 \\ 6 & 12 & 13 & 0 & 3 \end{bmatrix}.$$

Solution: Row reduce $\begin{bmatrix} A & \mathbf{0} \end{bmatrix}$:

$$\begin{bmatrix} 1 & 2 & 0 & 13 & 33 & 0 \\ 0 & 0 & 1 & -6 & -15 & 0 \end{bmatrix}$$

$x_1 = -2x_2 - 13x_4 - 33x_5$

$x_3 = 6x_4 + 15x_5$

x_2, x_4 and x_5 are free

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$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2x_2 - 13x_4 - 33x_5 \\ x_2 \\ 6x_4 + 15x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -13 \\ 0 \\ 6 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -33 \\ 0 \\ 15 \\ 0 \\ 1 \end{bmatrix}$$

$\uparrow$ $\uparrow$ $\uparrow$

u **v** **w**

Therefore $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a spanning set for Nul A. In the last section we observed that this set is linearly independent. Therefore $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a basis for Nul A. The technique used here always provides a linearly independent set.

§ 4.3 Linearly Independent Sets; Bases

Theorem 6

The pivot columns of a matrix A form a basis for Col A.

EXAMPLE: Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -2 \\ -4 \\ 6 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$.

Find a basis for Span$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

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§ 4.3 Linearly Independent Sets; Bases

Solution: Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ -3 & 6 & 9 \end{bmatrix}$ and note that Col $A = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

By row reduction, $A \sim \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Therefore a basis for $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} \right\}.$$

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§ 4.3 Linearly Independent Sets; Bases

- 5. Two View of a Basis

$$\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix}\right\}$$

线性无关，但不能生成$\mathbb{R}^3$

$$\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix},\begin{bmatrix}4\\5\\6\end{bmatrix}\right\}$$

$\mathbb{R}^3$的一个基

$$\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix},\begin{bmatrix}4\\5\\6\end{bmatrix},\begin{bmatrix}7\\8\\9\end{bmatrix}\right\}$$

生成$\mathbb{R}^3$，但线性相关

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§ 4.4 Coordinate System

- 1. Coordinate System

Theorem 7 The Unique Representation Theorem

Let $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis for a vector space V. Then for each $\mathbf{x}$ in V, there exists a unique set of scalars $c_1, \ldots, c_n$ such that

$$\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n.$$

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§ 4.4 Coordinate System

- Definition:

Suppose $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ is a basis for a vector space V and **x** is in V. The **coordinates of x relative to the basis β** (or the β – **coordinates of x**) are the weights $c_1, \ldots, c_n$ such that $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$.

In this case, the vector in $\mathbf{R}^n$

$$[\mathbf{x}]_\beta = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

is called the **coordinate vector of x** (**relative to β**), or the **β – coordinate vector of x**.

- **Coordinates in R^n**

- Let $b_1=\begin{bmatrix}2\\1\end{bmatrix}, b_2=\begin{bmatrix}-1\\1\end{bmatrix}, x=\begin{bmatrix}4\\5\end{bmatrix}, B=\{b_1,b_2\}$

  Find the coordinate vector $[\mathbf{x}]_\beta$ of **x** relative to **β**

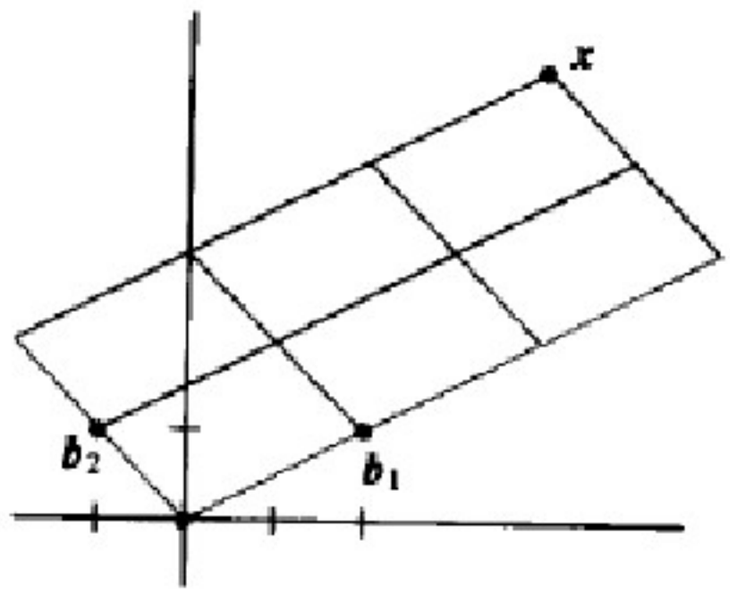
  Solution： The **β**-coordinate c_1, c_2 of **x** satisfy

  Thus $c_1\begin{bmatrix}2\\1\end{bmatrix}+c_2\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}4\\5\end{bmatrix}$ or $\begin{bmatrix}2&-1\\1&1\end{bmatrix}\begin{bmatrix}c_1\\c_2\end{bmatrix}=\begin{bmatrix}4\\5\end{bmatrix}$

  b_1 b_2 x b_1 b_2 x

  So $c_1=3, c_2=2$

  $\boldsymbol{x}=3\boldsymbol{b}_1+2\boldsymbol{b}_2$ $[\boldsymbol{x}]_B=\begin{bmatrix}c_1\\c_2\end{bmatrix}=\begin{bmatrix}3\\2\end{bmatrix}$



For a basis $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$, let

$$P_\beta = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{bmatrix} \text{ and } [\mathbf{x}]_\beta = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Then $\mathbf{x} = P_\beta [\mathbf{x}]_\beta$.

We call P_β the **change-of-coordinates matrix** from β to the standard basis in $\mathbf{R}^n$. Then

$$[\mathbf{x}]_\beta = P_\beta^{-1}\mathbf{x}$$

and therefore P_β^{-1} is a **change-of-coordinates matrix** from the standard basis in $\mathbf{R}^n$ to the basis β.

§ 4.4 Coordinate System

- **The Coordinate Mapping**

Coordinate mappings allow us to introduce coordinate systems for unfamiliar vector spaces.

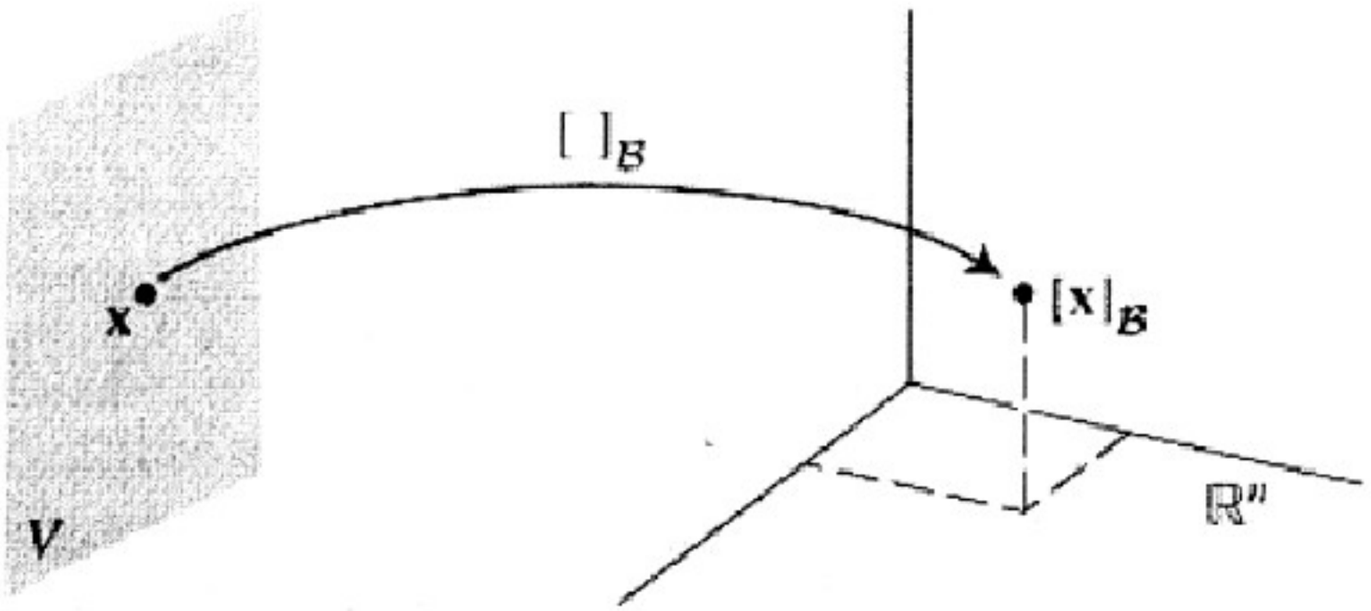


FIGURE 5 The coordinate mapping from V onto $\mathbb{R}^n$.

§ 4.4 Coordinate System

Theorem 8

Let $\beta = \{b_1,\ldots,b_n\}$ be a basis for a vector space V. Then the coordinate mapping $x \mapsto [x]_\beta$ is a one-to-one linear transformation from V onto R^n.

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EXAMPLE: Use coordinate vectors to determine if $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ is a linearly independent set, where $\mathbf{p}_1 = 1 - t$, $\mathbf{p}_2 = 2 - t + t^2$, and $\mathbf{p}_3 = 2t + 3t^2$.

Solution: The standard basis set for $\mathbf{P}_2$ is $\beta = \{1, t, t^2\}$. So

$$[\mathbf{p}_1]_\beta = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, [\mathbf{p}_2]_\beta = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, [\mathbf{p}_3]_\beta = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$$

Then

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & -1 & 2 \\ 0 & 1 & 3 \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

By the IMT, $\{[\mathbf{p}_1]_\beta, [\mathbf{p}_2]_\beta, [\mathbf{p}_3]_\beta\}$ is linearly **independent** and therefore $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ is linearly **independent**

§ 4.5 The Dimension of a Vector Space

- 1. Dimension

THEOREM 9

If a vector space V has a basis $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$, then any set in V containing more than n vectors must be linearly dependent.

THEOREM 10

If a vector space V has a basis of n vectors, then every basis of V must consist of n vectors.

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DEFINITION

If V is spanned by a finite set, then V is said to be **finite-dimensional**, and the **dimension** of V, written as dim V, is the number of vectors in a basis for V. The dimension of the zero vector space $\{\mathbf{0}\}$ is defined to be 0. If V is not spanned by a finite set, then V is said to be **infinite-dimensional**.

EXAMPLE: The standard basis for $\mathbf{P}_3$ is $\{\ 1, t, t^2, t^3, \quad\ \}$. So $\dim \mathbf{P}_3 =$ 4 .

In general, $\dim \mathbf{P}_n = n + 1$.

EXAMPLE: The standard basis for $\mathbf{R}^n$ is $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ where $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are the columns of I_n. So, for example, dim $\mathbf{R}^3 = 3$.

§ 4.5 The Dimension of a Vector Space

EXAMPLE: Find a basis and the dimension of the subspace

$$W = \left\{ \begin{bmatrix} a+b+2c \\ 2a+2b+4c+d \\ b+c+d \\ 3a+3c+d \end{bmatrix} : a,b,c,d \text{ are real} \right\}.$$

Solution: Since

$$\begin{bmatrix} a+b+2c \\ 2a+2b+4c+d \\ b+c+d \\ 3a+3c+d \end{bmatrix} = a\begin{bmatrix} 1 \\ 2 \\ 0 \\ 3 \end{bmatrix} + b\begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} + c\begin{bmatrix} 2 \\ 4 \\ 1 \\ 3 \end{bmatrix} + d\begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix},$$

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§ 4.5 The Dimension of a Vector Space

$$W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$$

where $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2 \\ 4 \\ 1 \\ 3 \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$

- Note that $\mathbf{v}_3$ is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$, so by the Spanning Set Theorem, we may discard $\mathbf{v}_3$.
- $\mathbf{v}_4$ is not a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. So $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is a basis for W. Also, dim $W =$ 3 .

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§ 4.5 The Dimension of a Vector Space

- 2. Subspaces of a Finite-Dimensional Space

THEOREM 11

Let H be a subspace of a finite-dimensional vector space V. Any linearly independent set in H can be expanded, if necessary, to a basis for H. Also, H is finite-dimensional and

$$\dim H \leq \dim V.$$

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§ 4.5 The Dimension of a Vector Space

EXAMPLE: Let $H = \text{span}\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$. Then H is a subspace of $\mathbf{R}^3$ and $\dim H < \dim \mathbf{R}^3$.

We could expand the spanning set $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$ to $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ to form a basis for $\mathbf{R}^3$.

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§ 4.5 The Dimension of a Vector Space

THEOREM 12 THE BASIS THEOREM

Let V be a $p-$ dimensional vector space, $p \geq 1$. Any linearly independent set of exactly p vectors in V is automatically a basis for V. Any set of exactly p vectors that spans V is automatically a basis for V.

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§ 4.5 The Dimension of a Vector Space

EXAMPLE: Show that $\{t, 1-t, 1+t-t^2\}$ is a basis for $\mathbf{P}_2$.

Solution: Let $\mathbf{v}_1 = t, \mathbf{v}_2 = 1-t, \mathbf{v}_3 = 1+t-t^2$ and $\beta = \{1, t, t^2\}$.

Corresponding coordinate vectors

$$[\mathbf{v}_1]_\beta = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, [\mathbf{v}_2]_\beta = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, [\mathbf{v}_3]_\beta = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

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§ 4.5 The Dimension of a Vector Space

$[\mathbf{v}_2]_\beta$ is not a multiple of $[\mathbf{v}_1]_\beta$

$[\mathbf{v}_3]_\beta$ is not a linear combination of $[\mathbf{v}_1]_\beta$ and $[\mathbf{v}_2]_\beta$

$\Rightarrow \{[\mathbf{v}_1]_\beta, [\mathbf{v}_2]_\beta, [\mathbf{v}_3]_\beta\}$ is linearly independent and therefore $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is also linearly independent.

Since dim $\mathbf{P}_2 = 3$, $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis for $\mathbf{P}_2$ according to The Basis Theorem.

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§ 4.5 The Dimension of a Vector Space

- 3. The Dimensions of Nul A and Col A

$$\boxed{\dim \text{Col}\, A = \text{number of pivot columns of } A}$$

$$\boxed{\dim \text{Nul}\, A = \text{number of free variables of } A}.$$

EXAMPLE: Suppose $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 7 & 8 \end{bmatrix}$. Find dim Col A and dim Nul A.

Solution

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 7 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

So $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \end{bmatrix} \right\}$ is a basis for Col A and dim Col $A = 2$.

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§ 4.5 The Dimension of a Vector Space

Now solve $A\mathbf{x} = \mathbf{0}$ by row-reducing the corresponding augmented matrix. Then we arrive at

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 0 \\ 2 & 4 & 7 & 8 & 0 \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 & 2 & 0 & 4 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$x_1 = -2x_2 - 4x_4$$

$$x_3 = 0$$

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§ 4.5 The Dimension of a Vector Space

and

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -4 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

So $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for Nul A and dim Nul $A = 2$.

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§ 4.6 Rank

1. The Row Space

The set of all linear combinations of the row vectors of a matrix A is called the **row space** of A and is denoted by Row A.

EXAMPLE: Let

$$A = \begin{bmatrix} -1 & 2 & 3 & 6 \\ 2 & -5 & -6 & -12 \\ 1 & -3 & -3 & -6 \end{bmatrix} \quad \text{and} \quad \begin{aligned} \mathbf{r}_1 &= (-1,\ 2,\ 3,\ 6) \\ \mathbf{r}_2 &= (2,-5,-6,-12) \\ \mathbf{r}_3 &= (1,-3,-3,-6) \end{aligned} .$$

$$\text{Row } A = \text{Span}\{\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3\} \text{ (a subspace of } \mathbf{R}^4)$$

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While it is natural to express row vectors horizontally, they can also be written as column vectors if it is more convenient. Therefore

$$\boxed{\text{Col } A^T = \text{Row } A}.$$

When we use row operations to reduce matrix A to matrix B, we are taking linear combinations of the rows of A to come up with B. We could reverse this process and use row operations on B to get back to A. Because of this, the row space of A equals the row space of B.

§ 4.6 Rank

THEOREM 13

If two matrices A and B are row equivalent, then their row spaces are the same. If B is in echelon form, the nonzero rows of B form a basis for the row space of A as well as B.

- Example： Find bases for the row space, the column space, and the null space of the matrix

$$A=\begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix}$$

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§ 4.6 Rank

- Solution: Row reduce A to an echelon form:

$$A \sim B = \begin{bmatrix} 1 & 3 & -5 & 1 & 5 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & -4 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Basis for Row A : $\{(1,3,-5,1,5),(0,1,-2,2,-7),(0,0,0,-4,20)\}$

Basis for Col A :

$$\left\{\begin{bmatrix} -2 \\ 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 11 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 7 \\ 5 \end{bmatrix}\right\}$$

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To find Nul A, solve $A\mathbf{x} = \mathbf{0}$ first:

$$A \sim B \sim C = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{aligned} x_1 + x_3 + x_5 &= 0 \\ x_2 - 2x_3 + 3x_5 &= 0 \\ x_4 - 5x_5 &= 0 \end{aligned}$$

$x_1 = -x_3 - x_5, x_2 = 2x_3 - 3x_5, x_4 = 5x_5$，$x_3$ 和 x_5 为自由变量，

Basis for Nul A : $\left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 1 \end{bmatrix} \right\}$

§ 4.6 Rank

Note the following:

$$\dim \text{Col } A = \text{\# of pivots of } A = \text{\# of nonzero rows in } B$$

$$= \dim \text{Row } A.$$

$$\dim \text{Nul } A = \text{\# of free variables} = \text{\# of nonpivot columns of } A.$$

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§ 4.6 Rank

- The Rank Theorem

DEFINITION

The **rank** of A is the dimension of the column space of A.

$$\boxed{\text{rank } A = \dim \text{Col } A = \text{\# of pivot columns of } A = \dim \text{Row } A}.$$

$$\underbrace{\text{rank } A}_{\updownarrow} + \underbrace{\dim \text{Nul } A}_{\updownarrow} = \underbrace{n}_{\updownarrow}$$

$$\left\{\begin{array}{c}\text{\# of pivot}\\ \text{columns}\\ \text{of } A\end{array}\right\} \quad \left\{\begin{array}{c}\text{\# of nonpivot}\\ \text{columns}\\ \text{of } A\end{array}\right\} \quad \left\{\begin{array}{c}\text{\# of}\\ \text{columns}\\ \text{of } A\end{array}\right\}$$

THEOREM 14 THE RANK THEOREM

The dimensions of the column space and the row space of an $m \times n$ matrix A are equal. This common dimension, the rank of A, also equals the number of pivot positions in A and satisfies the equation

$$\text{rank } A + \dim \text{Nul } A = n.$$

Since Row A = Col A^T,

$$\boxed{\text{rank } A = \text{rank } A^T}.$$

§ 4.6 Rank

3. Rank and the Invertible Matrix Theorem

THE INVERTIBLE MATRIX THEOREM (continued)

Let A be a square $n \times n$ matrix. The the following statements are equivalent:

m. The columns of A form a basis for $\mathbf{R}^n$

n. Col $A = \mathbf{R}^n$

o. dim Col $A = n$

p. rank $A = n$

q. Nul $A = \{\mathbf{0}\}$

r. dim Nul $A = 0$

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